

# SECTION 7.E

## Above & Beyond: Riemann Sums and Trapezoidal Approximation

### Learning Objectives

I can approximate the area under a curve using left and right Riemann sums from graphs and tables.

I can apply trapezoidal sums to approximate area.

I can compare approximation methods and justify which provides a better estimate.

I can model real-world situations involving accumulation.

### Real-World Application

Many real-world quantities cannot be measured exactly.

- Total rainfall during a storm.
- Distance traveled when speed changes continuously.
- Total energy used over time.
- Accumulated medication levels in the bloodstream.

Suppose a car's velocity (in miles per hour) is recorded every 2 seconds.

We want to estimate the total distance traveled over 10 seconds.

Because velocity is changing, we cannot multiply by a single constant speed.

Instead, we approximate the area under the velocity curve.

The area under a velocity–time graph represents **distance traveled**.

This is the foundation of calculus.

### Key Vocabulary

**Riemann Sum:** Approximation of area using rectangles.

**Left Riemann Sum:** Uses left endpoints.

**Right Riemann Sum:** Uses right endpoints.

**Trapezoidal Rule:** Uses trapezoids to approximate area.

### Core Concepts

Increasing  $\rightarrow$  Left underestimates, Right overestimates.

Decreasing  $\rightarrow$  Left overestimates, Right underestimates.

Trapezoidal averages endpoint behavior.

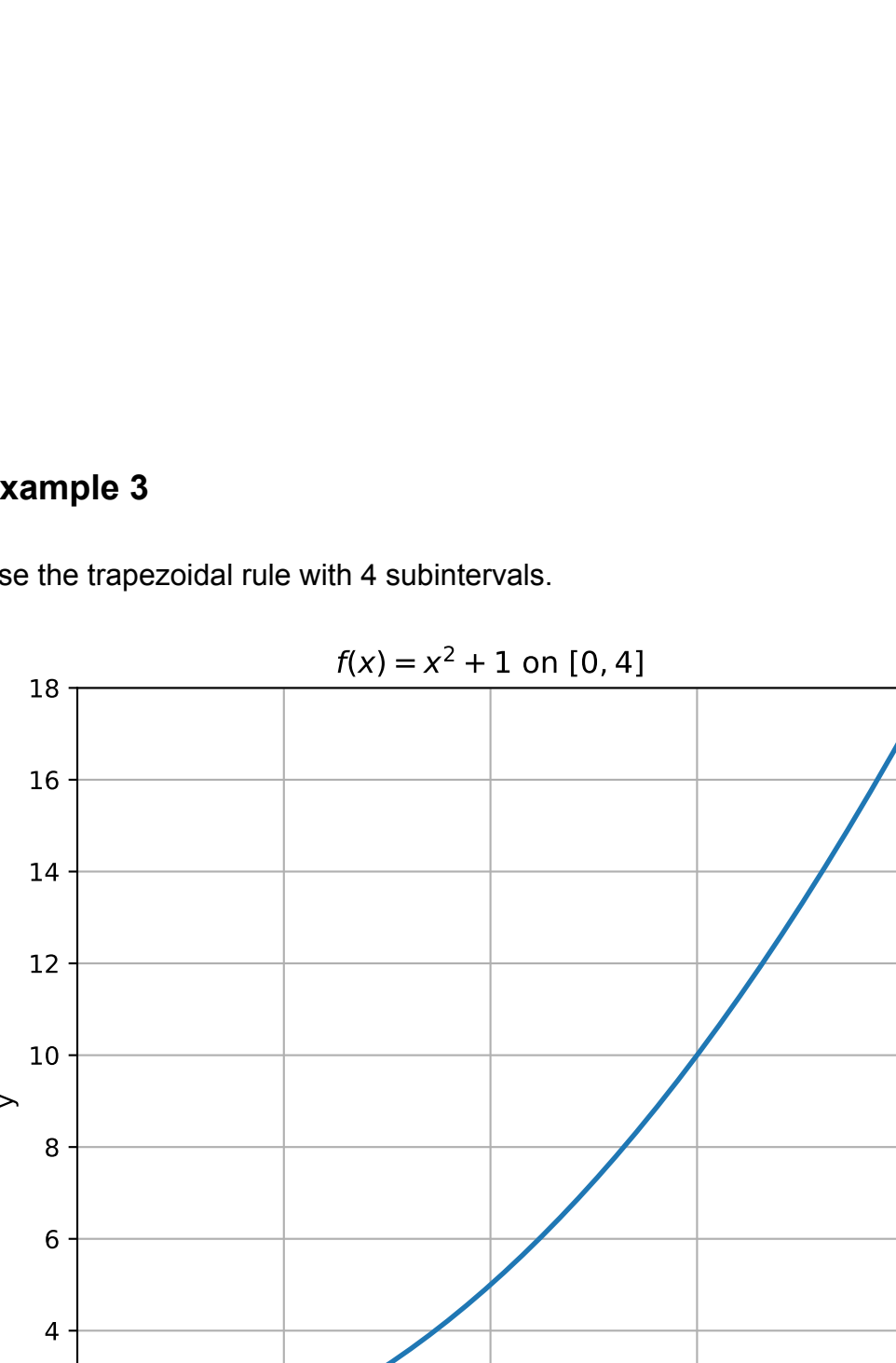
### Structure Insight

- We estimate area numerically.
- Units matter.
- More subintervals improve accuracy.
- We justify reasoning, not just compute.

### Examples

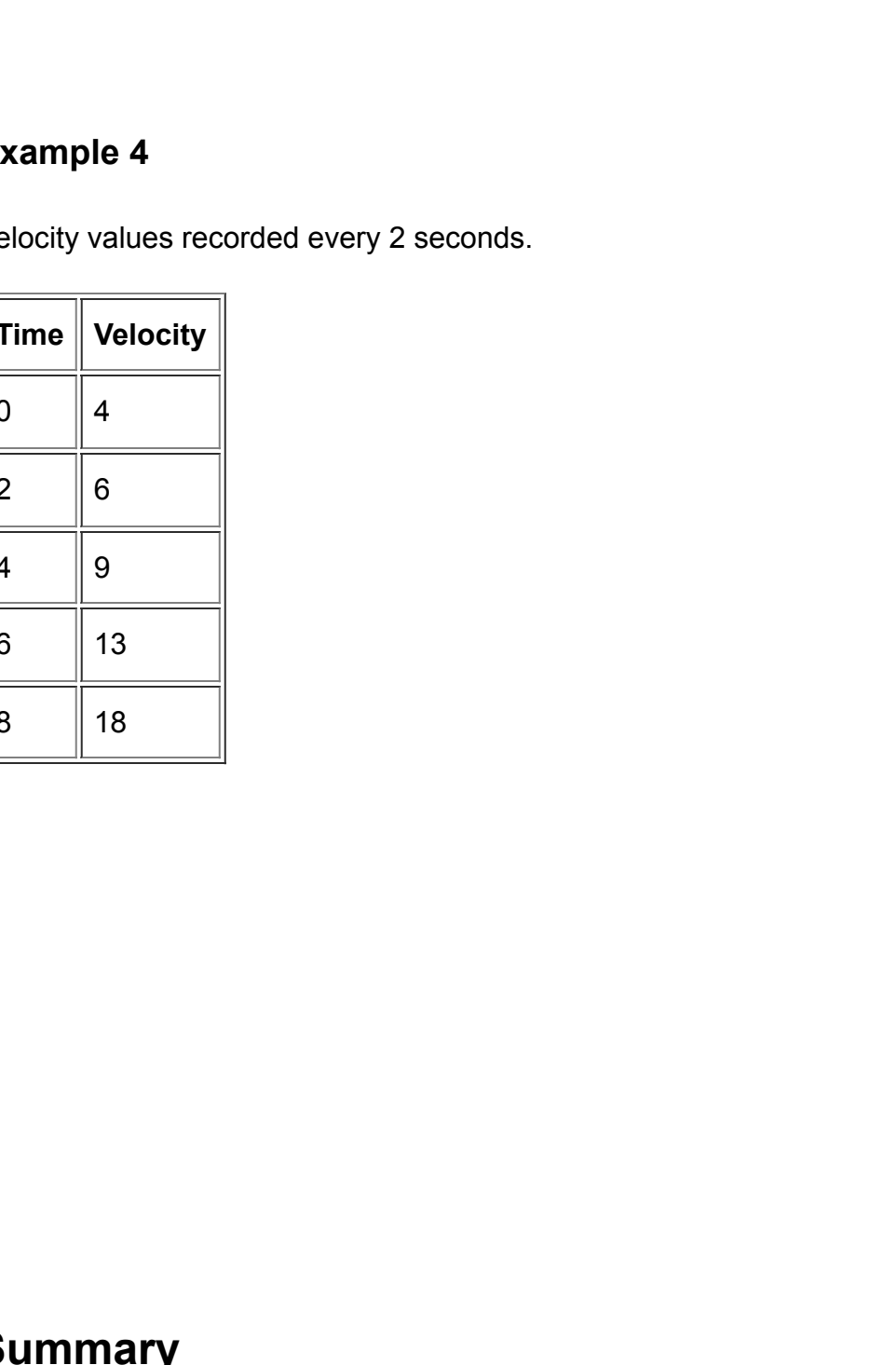
#### Example 1

Use a left Riemann sum with 4 subintervals to approximate area under  $f(x) = x^2 + 1$  on  $[0, 4]$ .



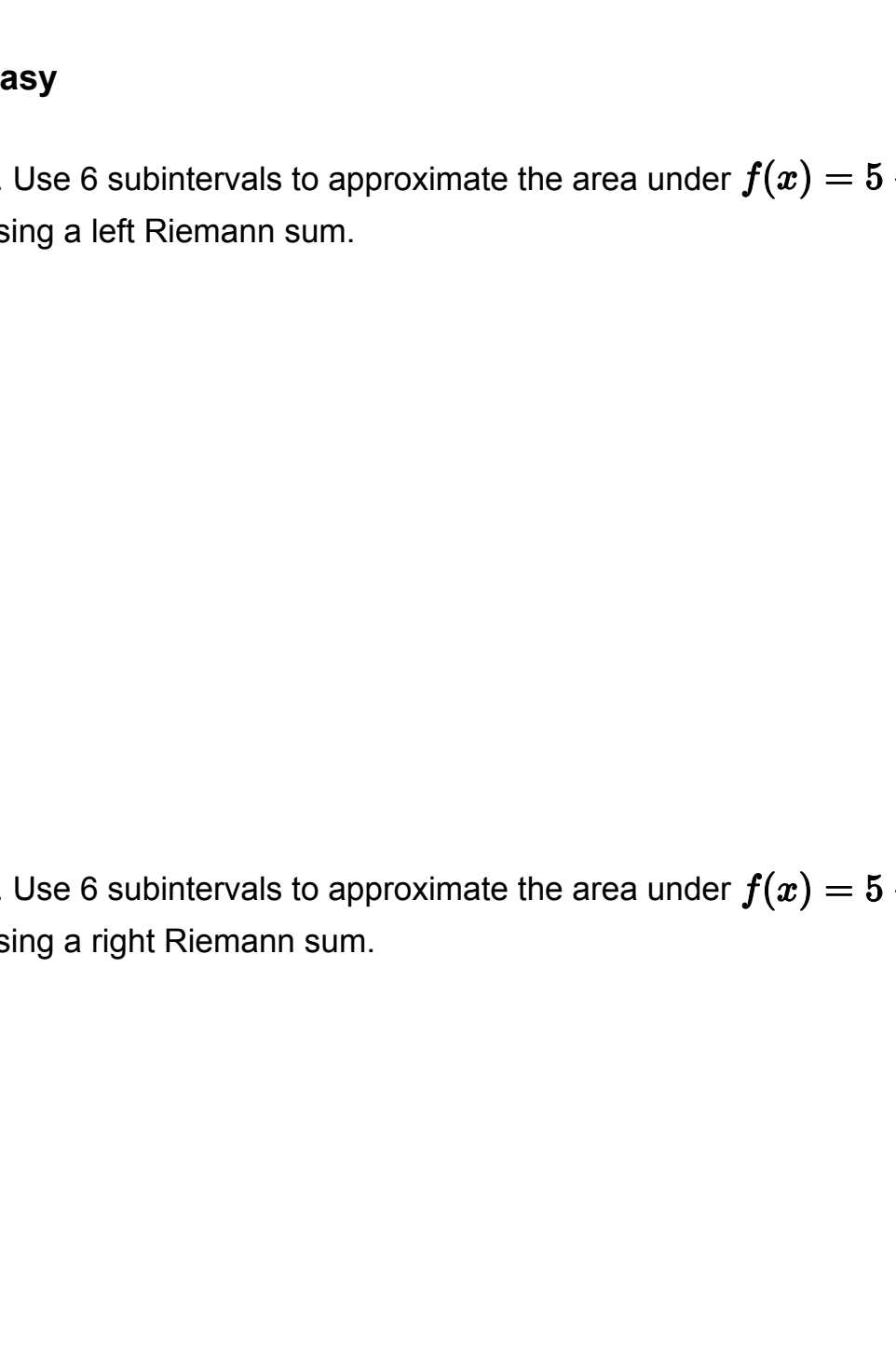
#### Example 2

Use a right Riemann sum with 4 subintervals for the same function.



#### Example 3

Use the trapezoidal rule with 4 subintervals.



#### Example 4

Velocity values recorded every 2 seconds.

| Time | Velocity |
|------|----------|
| 0    | 4        |
| 2    | 6        |
| 4    | 9        |
| 6    | 13       |
| 8    | 18       |

### Summary

- Riemann sums approximate area.
- Left and right behave differently.
- Trapezoidal improves accuracy.
- Increasing subintervals improves precision.

### Common Mistakes

- Forgetting to multiply by  $\Delta x$ .
- Mixing left and right endpoints.
- Ignoring units.
- Failing to interpret over/under estimates.

### Practice Problems

#### Easy

1. Use 6 subintervals to approximate the area under  $f(x) = 5 - x^2$  on  $[-3, 3]$  using a left Riemann sum.

2. Use 6 subintervals to approximate the area under  $f(x) = 5 - x^2$  on  $[-3, 3]$  using a right Riemann sum.

3. Determine whether the left Riemann sum in Problem 1 is an overestimate or underestimate. Justify using function behavior.

4. Compute  $\Delta x$  for 6 subintervals on  $[-3, 3]$ . Explain what it represents geometrically.

#### Medium

5. Use 4 subintervals to approximate the area under  $f(x) = 2^x + 3$  on  $[1, 3]$  using a left Riemann sum.

6. Use 4 subintervals to approximate the area under  $f(x) = 2^x + 3$  on  $[1, 3]$  using a right Riemann sum.

7. Use 4 subintervals to approximate the area under  $f(x) = 2^x + 3$  on  $[1, 3]$  using the trapezoidal rule.

8. Compare your answers from 5–7. Which method gives the largest estimate? Explain why.

#### Hard

9. Use 10 subintervals to approximate the area under  $f(x) = x^2 + 4$  on  $[1, 6]$  using a left Riemann sum.

10. Use 10 subintervals to approximate the area under  $f(x) = x^2 + 4$  on  $[1, 6]$  using a right Riemann sum.

11. Use 10 subintervals to approximate the area under  $f(x) = x^2 + 4$  on  $[1, 6]$  using the trapezoidal rule.

12. Compare Problems 9–11. Which approximation should be closest to the true value? Explain using concavity.

### Challenge

13. A velocity function is increasing and concave up on an interval. Predict which method (left, right, trapezoidal) will give the smallest estimate. Justify.

14. A function is decreasing and concave down. Predict which method will give the largest estimate. Explain.

15. Without calculating exact values, explain how doubling the number of subintervals affects approximation error.

### Justify

16. Explain why increasing the number of subintervals improves approximation accuracy.

17. Justify why the trapezoidal rule is often more accurate than a left or right Riemann sum.

18. Explain how Riemann sums connect to the definition of a definite integral in calculus.

### Solutions

#### Easy

1.  $\Delta x = \frac{3-(-3)}{6} = 1$  Left endpoints:  $-3, -2, -1, 0, 1, 2$   $f(x) = 5 - x^2$  Values:  $f(-3) = -4$   $f(-2) = 1$   $f(-1) = 4$   $f(0) = 5$   $f(1) = 4$   $f(2) = 1$  Left sum:  $L = 1(-4 + 1 + 4 + 5 + 4 + 1) = 11$

2. Right endpoints:  $-2, -1, 0, 1, 2, 3$  Values:  $1, 4, 5, 4, 1, -4$  Right sum:  $R = 1(1 + 4 + 5 + 4 + 1 - 4) = 11$

3. Because  $5 - x^2$  is symmetric about the y-axis, left and right sums balance. Neither strictly over- nor underestimates over symmetric interval.

4.  $\Delta x = 1$  This represents the width of each rectangle.

#### Medium

5.  $\Delta x = \frac{3-1}{4} = 0.5$  Left endpoints:  $1, 1.5, 2, 2.5$   $f(x) = 2^x + 3$   $f(1) = 5$   $f(1.5) = 2^{1.5} + 3 \approx 5.828$   $f(2) = 7$   $f(2.5) = 2^{2.5} + 3 \approx 8.657$   $L \approx 0.5(5 + 5.828 + 7 + 8.657) = 13.2425$

6. Right endpoints:  $1.5, 2, 2.5, 3$   $f(3) = 11$   $R \approx 0.5(5.828 + 7 + 8.657 + 11) = 16.7425$

7.  $T = \frac{L+R}{2} \approx \frac{13.2425+16.7425}{2} = 14.9925$

8. Function increasing and concave up. Left underestimates, right overestimates, trapezoidal slightly overestimates but closest.

#### Hard

9.  $\Delta x = \frac{6-1}{10} = 0.5$  Left endpoints:  $1, 1.5, 2, \dots, 5.5$  Compute each and multiply by 0.5 Left sum  $\approx 108.125$

10. Right endpoints:  $1.5, 2, \dots, 6$  Right sum  $\approx 133.125$

11.  $T = \frac{108.125+133.125}{2} = 120.625$

12. Since  $x^2 + 4$  is increasing and concave up: Left underestimates, right overestimates, trapezoidal closest but slightly over.

### Challenge

13. Increasing & concave up  $\rightarrow$  Left gives smallest estimate.

14. Decreasing & concave down  $\rightarrow$  Left gives largest estimate.

15. Doubling subintervals reduces error because rectangles better approximate curvature.

16. Smaller  $\Delta x$  reduces maximum gap between curve and rectangles.

17. Trapezoidal balances left and right error, canceling part of over/under estimation.

18. As  $\Delta x \rightarrow 0$ , Riemann sums converge to the definite integral.